



Animated Ambient Logo Background — Design Specification

Site-wide shining NT-monogram watermark over drifting aurora, theme-aware

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1 Overview

Replace the current per-section static backdrops with a single **site-wide animated ambient background**: a large, low-opacity **NT monogram watermark** that gently drifts and “shines”, floating over slow-moving aurora glows and a faint blueprint grid. The effect is **theme-aware** (distinct palettes for dark and light) and reacts subtly to the mouse for depth. It renders **once**, behind all public pages, and is suppressed on `/admin`.

1.1 Goals

- A premium, branded atmosphere that reinforces the NurvexThink mark on every public page.
- A tasteful “shine” that sweeps across the logo watermark, clipped to the letterforms.
- Smooth, GPU-friendly motion that never harms readability or Core Web Vitals.
- Fully theme-aware, honoring `prefers-reduced-motion`.

1.2 Non-goals

- No canvas / WebGL / shader background (the hero already owns one GPU context).
- No per-page configuration or CMS control of the background.
- Not shown on admin pages (kept clean and functional).

2 Locked decisions

Decision	Choice
Scope	Site-wide — one fixed layer behind all public pages; excluded on <code>/admin*</code>
Style	Layered atmosphere — centered NT monogram + drifting aurora + faint grid + shine
Motion	Ambient + mouse parallax (parallax disabled on touch and reduced-motion)
Intensity	Balanced — clearly present, always behind content; watermark ~6–10% opacity
Technique	Pure CSS transforms + inline SVG (no canvas / WebGL)

3 Approach

Chosen: **CSS transforms + inline SVG**, animating only `transform` and `opacity` so all motion runs on the compositor. Rejected alternatives: a WebGL / canvas shader (a second always-on GPU context site-wide — overkill, worse battery, conflicts with the “lazy-load 3D so it never blocks paint” rule) and animated SVG filters / `feTurbulence` (CPU-heavy, hard to theme).

4 Architecture

One client component, mounted once in the root layout.

Unit	Responsibility
<code>components/ambient-background.tsx</code> ("use client")	The fixed inset-0 -z-10 pointer-events-none layer, aria-hidden. Owns the mouse-parallax effect and visibility rules. Returns null on /admin*.
<code>components/nt-mark.tsx</code>	The traced NT monogram as a reusable inline SVG, tintable via <code>currentColor</code> / CSS vars. Swappable for an official SVG later.
<code>app/globals.css</code>	New keyframes (aurora-drift, gleam-sweep, breathe) and <code>.ambient-*</code> classes, theme-aware through existing tokens.
<code>app/layout.tsx</code>	Mounts <code><AmbientBackground /></code> as the first child of <code><body></code> .

4.1 Layering (back to front)

1. **Base** — existing `--background` token (untouched; painted by `<body>`).
2. **Aurora** — 2–3 large soft radial-gradient blobs, slowly drifting + breathing.
3. **Grid** — the existing faint blueprint grid, full-page, edge-masked.
4. **Monogram watermark** — large, centered NT, low opacity, gently breathing.
5. **Shine** — a diagonal light band sweeping across the monogram every ~6–8 s, **clipped to the letter shapes** via an SVG `clipPath` so the gleam only rides the logo.

The ambient layer lives at `-z-10` as a child of `<body>`. Per CSS paint order, a negative `z-index` child paints *above* the body's own `bg-background` fill but *below* all in-flow content — so it is visible in transparent regions and correctly hidden behind opaque `bg-card` surfaces.

4.2 Mouse parallax

A single `pointermove` listener (throttled with `requestAnimationFrame`) writes normalized `--mx` / `--my` (`-1...1`) CSS variables on the layer root. Each layer translates a different magnitude toward the cursor (monogram least, aurora most) for depth. The listener is **not attached** when `(hover: none)` (touch) or `prefers-reduced-motion: reduce`.

5 Theme-aware palette

Layer	Dark	Light
Aurora	indigo #5C7CFA + navy #1E2A44	indigo #8AAOFF + cool blue, softer
Monogram fill	silver #C0C5CE → #E8EBF0, ~8%	navy #1E2A44 → #2D3A5C, low opacity
Gleam	bright silver / white	white / indigo
Grid	existing <code>--grid-line</code> (faint silver)	existing <code>--grid-line</code> (faint navy)

Colors are driven by existing CSS custom properties, so the `.dark` class switch flips the whole background automatically.

6 Integration adjustments

- **Home hero** (`src/app/page.tsx`): remove the now-redundant local `.bg-grid` (the global grid covers it); keep the 3D crystal and a subtle local `.bg-glow` for focal emphasis at the top of the hero.
- **Admin** (`src/app/admin/(panel)/`): background suppressed via the `usePathname` check in `ambient-background.tsx`.

7 Performance & accessibility

- Animate **only** transform / opacity; will-change applied sparingly and scoped.
- `pointer-events: none` and `aria-hidden="true"` on the whole layer.
- Full prefers-reduced-motion support: the existing global guard in `globals.css` freezes animation; the parallax listener additionally self-disables.
- No layout thrash, no per-frame React state, no new dependencies.

8 Testing

- Component render test (`ambient-background.test.tsx`) matching the existing `navbar.test.tsx` pattern: asserts the layer renders `aria-hidden`, and renders nothing on an `/admin` path (mocked `usePathname`).
- Keep the green bar: `npm run build`, `npm run typecheck`, `npm run lint`, `npm test` all pass.
- Manual: verify the shine sweep, aurora drift, and mouse parallax in both themes; confirm the background disappears on `/admin` and that content readability is unaffected.

9 Asset note

The provided `public/logo.jpeg` is a raster JPEG on a black background — it cannot be tinted or masked. We therefore **trace a clean transparent SVG** of the NT monogram for `nt-mark.tsx`. It will closely match the mark; an official SVG can be dropped in later without other changes.

10 Build order

1. Add `nt-mark.tsx` (traced SVG monogram).
2. Add keyframes + `.ambient-*` classes to `globals.css` (theme-aware).
3. Build `ambient-background.tsx` (layers + parallax + `/admin` suppression).
4. Mount in `layout.tsx`; trim the home hero's redundant grid.
5. Add the render test; run the full green-bar suite.
6. Visual QA in both themes; tune opacity / timing.